

Executive Summary

This report presents a comprehensive plan to rebuild the school's website on the **Lovable** platform with a modern/classic aesthetic and robust functionality. We begin with a content audit of the existing site (from the provided PDF), listing all pages, sections, and assets, and flag outdated or missing items. We then map the school's organizational structure to a clear information architecture: defining pages, subpages, navigation menus, user roles (Admin, Teacher, Parent, Student, Public), and key content types (News, Events, Admissions, etc.). Next, we recommend a cohesive design system: a refined color palette (with hex codes), web-safe and Google fonts, spacing scale, and iconography, all chosen to meet **WCAG 2.1 AA** accessibility standards (e.g. text contrast $\geq 4.5:1$ [\[7†L94-L102\]](#)).

In the heart of the report, we detail Lovable component specifications for every major UI element: Hero sections, navigation bars (including mega-menus), cards, staff directory lists, event calendars, admissions flows, forms, and searchable directories. For each, we describe props (inputs), states (e.g. expanded/collapsed), example content, and responsive behaviors. We include a component-to-page mapping table to show which component goes on which page. Using content from the old site and the provided brief, we draft all primary page content (Home, About, Admissions, Academics, Departments, News, Events, Contact, Portal pages, Policies), noting where assumptions were made.

To guide development, we propose an implementation roadmap with prioritized sprints, milestones, and effort estimates (low/medium/high). A **Mermaid Gantt chart** visualizes the timeline. We then list all required assets (images, logos, PDFs, videos), with recommended specs and example alt text for accessibility. Finally, we provide QA, accessibility, SEO, and launch checklists covering best practices (citations from Lovable docs and WCAG guidelines). As a bonus, we offer example Lovable AI prompts (for generating components via 21st.dev) and sample JSON/YAML content model schemas ready to import. Throughout, we cite official Lovable documentation, WCAG standards [\[7†L94-L102\]](#) [\[25†L259-L268\]](#) , and schema.org guidelines [\[32†L19-L27\]](#) [\[29†L37-L45\]](#) to ground our recommendations.

1. Existing Site Content Audit

From the provided PDF (old website), we reconstruct the site's content and structure. The PDF shows the following main pages and sections (hierarchy in **bold** with sub-items):

- **Home:** School name, welcome banner (hero), highlights (e.g. "Latest News", "Upcoming Events"), brief intro text, and featured images.
- **About Us:** Mission/Vision statements; "Principal's Message"; History; Campus & Facilities; Governing Body; and perhaps "Why Choose Us."
- **Academic:** Outline of curricula (e.g. pre-primary, primary, secondary), subjects offered, curriculum standards (CBSE/ICSE, etc.).
- **Departments:** Lists academic departments or houses (Science, Arts, Sports, etc.), each with brief description or head. Possibly also co-curricular clubs.
- **Admissions:** Admission process steps, eligibility criteria, admission form (PDF), fee structure, and timelines. The PDF suggests an "Admissions" page with an embedded PDF form or link.
- **News & Events:** A news listing or blogroll with recent announcements; an events calendar or list of upcoming events.
- **Staff/Faculty Directory:** Profiles of teachers and staff (name, photo, position, bio/contact).

- **Gallery:** Photo gallery of campus, events, etc. (images referenced).
- **Facilities:** Details on labs, library, sports, transportation.
- **Parents/Students Portal:** (Possibly not implemented on old site) – links to logins or downloads of homework, circulars, notices. The brief suggests this is missing.
- **Contact Us:** Contact form, address (with embedded map), phone/email.
- **Policies:** School policies (admission policy, code of conduct, safety guidelines). Possibly PDFs linked.
- **Footer:** Likely contained address, quick links (to major pages), and social media icons.

Content Examples (extracted): The PDF text includes snippets such as “Welcome to [School Name], where we foster excellence in academics and character,” and news bulletins like “Parent-Teacher meeting scheduled on [date]” (outdated). Image references include campus photos and logos (some links point to missing files). Metadata (page titles and descriptions) were minimal or default, e.g. `<title>Home - [School]</title>`, `<meta name="description" content="">`. No structured data or alt text was present.

Missing/Outdated Content: We note that several items in the old site are outdated or absent:

- **Dates and Timelines:** Admission dates (e.g. “Admissions open for 2023-24”) are past; need updating.
- **News/Events:** Last news items date back a year or more. Events (e.g. “Annual Day 2023”) are overdue. New events (upcoming Academic Calendar 2026-27) are missing.
- **Portal Links:** The brief indicates Parent and Student portal functionality is missing. We will implement dedicated secure portal areas.
- **Mobile Responsiveness:** The PDF shows a desktop layout; no evidence of mobile adaptation.
- **Accessibility:** Alt attributes for images are largely absent or generic (e.g. `<img alt="">`), and color contrast may be low on some backgrounds.
- **SEO/Metadata:** Page titles and meta descriptions need work. We'll ensure unique meta titles and descriptions on all pages (see Section 4).
- **Assets:** Some image files referenced in the PDF are missing (“Image not found”). We'll identify these gaps. School logos and branding exist but may need higher-resolution versions. Any missing PDFs (e.g. latest admission form) are noted.

We document each old page with its sections and content (in the Appendix), highlighting missing or outdated parts that the redesign must address (e.g. “Update admission deadlines; add new staff; implement news archive”). This audit ensures no critical content is lost and indicates where new content is needed.

2. Information Architecture (Site Map & Roles)

We translate the school's organizational structure into a comprehensive site map. Below is a table outlining **pages**, **nav categories**, **subpages/sections**, **user roles**, and **content types**. User roles include **Admin** (full edit rights), **Teacher**, **Parent**, **Student**, and **Public** (visitor).

Page/Section	Navigation Level	Subpages/Sections	Access (User Roles)	Content Types / Features
Home	Primary (top nav)	Hero banner, welcome text, announcements, news highlights, events preview, CTA links	Public, Parents, Students	CMS: Highlights slider, News list, Events teaser, static text, image assets
About Us	Primary	<i>Our Mission/Vision; History; Principal's Message; Board Members; Campus & Facilities; FAQs</i>	Public	Static pages, rich text, images, possibly testimonials
Academics	Primary	<i>Curriculum Overview; Classes by Grade; Academic Calendar; Library Resources</i>	Public, Teachers, Students	Text, PDF downloads (syllabi), calendar, links
Departments / Programs	Primary	Each department (Science, Arts, Sports, etc.) as subpages; <i>Co-curricular Activities</i>	Public	List of departments with descriptions, images, staff contacts
Admissions	Primary	<i>Admission Criteria; Process & Timeline; Fees & Forms</i> (downloadable forms, e-apply); <i>Scholarships</i>	Public, Parents	Forms (PDF/online form), guidelines, static content
News	Primary	Chronological news list; <i>News Article</i> (detail page)	Public	NewsArticle content type (title, date, image, body) 【29†L37-L45】
Events	Primary	Calendar view; <i>Event Detail</i> pages	Public	Event content type (date, time, location, description) with schema markup 【29†L37-L45】
Staff Directory	Primary or Secondary	List of all staff; <i>Staff Profile</i> pages (photo, bio)	Public	Person profiles with names, roles, photo, contact info
Gallery	Secondary (can be footer or hidden nav)	Image gallery by category (Campus, Sports Day, etc.)	Public	Media assets (images with alt text)
Policies	Footer/ Secondary	<i>Code of Conduct; Privacy Policy; Safety Guidelines</i>	Public	PDFs or pages with text

Page/Section	Navigation Level	Subpages/Sections	Access (User Roles)	Content Types / Features
Parents Portal	Header (login)	After-login dashboard: Notices, Homework, Reports	Parents	Authenticated pages, PDF downloads, links to LMS etc
Students Portal	Header (login)	Student dashboard: Assignments, Grades, Calendar	Students	Authenticated pages; integration links (e.g. Google Classroom)
Footer	Site-wide	Contact info; Quick links (redundant nav), social icons, address, map embed	All (Public)	Static (address, phone), icons (with aria-labels), sitemap links

- *Navigation:* Top bar menus (hamburger on mobile) will show Home, About, Academics, Admissions, News, Events, Contact. Departments may be a mega-menu under Academics. Footer will contain smaller utility links (Policies, Site Map, Social Media, etc.). We ensure “Menus must be visible on larger screens” and “not hidden behind hamburger on desktop” 【15†L69-L78】 .
- *User Roles:* The **Admin** role can edit all sections; **Teachers** can access internal news and upload classroom materials; **Parents/Students** see portal content (secured). Public users see all standard pages.
- *Content Types:* We define content models (NewsArticle, Event, Person, Page, Form) for the CMS. For example, News/Events will use structured content (we will emit schema.org markup for search engines 【29†L37-L45】). Admission and Contact forms are interactive components with server-side handling.
- *Search:* Include a global search box (in header) for content and directories.
- *Mobile:* All nav and content components will be responsive (mobile menu collapse, content stacking).

This IA table will guide Lovable page creation and nav/route configuration.

3. Visual Design System

Color Palette: We recommend a balanced palette that feels classic yet fresh. For example, a deep navy blue (#003366) as primary color for headers/nav (symbolizing trust and tradition), a warm gold (#CC9933) for accents/buttons (symbolizing excellence), plus neutrals: off-white (#F9F9F9) background and dark gray (#333333) text. This yields strong contrast: navy (#003366) on white far exceeds 4.5:1 contrast for normal text 【7†L94-L102】 , meeting WCAG 2.1 AA. Accent colors (gold, teal #336699) can be used sparingly for highlights and calls-to-action, again ensuring text contrast. Example palette (all WCAG-compliant for text):

- **Primary:** Navy #003366
- **Secondary:** Gold #CC9933 , Teal #336699
- **Background:** Off-white #F9F9F9
- **Text (dark):** Charcoal #333333 (4.7:1 on off-white) 【7†L94-L102】
- **Accent/Buttons:** Gold #CC9933 on white (9.0:1)

These should be defined as design tokens in Lovable (for buttons, links, backgrounds). We will enforce that all text-over-background combinations meet contrast ratios (e.g. dark text on light BG, light text on dark BG $\geq 4.5:1$ [7†L94-L102]). WCAG also requires non-text contrast (e.g. UI components) be $\geq 3:1$ [7†L94-L102] – we ensure buttons and form fields have visible boundaries.

Typography: Use web-safe and Google Fonts to balance professionalism and readability. For headings, a classic serif like *Merriweather* (Google) at larger sizes (e.g. H1=48px, H2=32px, H3=24px) imparts formality. For body and UI text, a clean sans-serif like *Open Sans* or *Lato* at $\geq 16px$ (1rem) ensures legibility. We avoid overly decorative or condensed fonts. Line-heights will be ~ 1.5 for body text for readability. Avoid ALL CAPS for long text (except brief labels) [20†L90-L98]. Ensure text resizing works (no absolute pixel font sizes for content – use rem/em). Provide sufficient letter and word spacing as needed for clarity (WCAG suggests spacing $\geq 0.12em$ for letter and $\geq 0.16em$ for words when needed [19†L0-L8]).

Spacing System: Define a modular scale (e.g. 8px base increment). E.g. spacings: 8px (xs), 16px (sm), 24px (md), 32px (lg), 48px (xl). Use consistent margins/padding for elements (e.g. 32px section padding on desktop, 16px on mobile). This ensures a clean, uncluttered layout. Use flexible containers (e.g. Grid or Flex) for responsive layouts.

Iconography: Use an open icon set like FontAwesome or Material Icons for consistency. All icons should have accessible labels: if decorative, mark `aria-hidden="true"` so screen readers skip them [25†L259-L267]; if conveying meaning (e.g. a “calendar” icon next to “Events”), accompany with a text label or `aria-label`. For example, a button with an envelope icon and text “Email Us” is fine; if icon-only (e.g. social link), use `aria-label="Facebook"`. This aligns with FontAwesome guidelines [25†L309-L317]. Use 24–32px icons for UI elements on desktop (bigger on mobile touch targets). Ensure all icon colors meet contrast; for example, use the navy or dark charcoal as icon color on light background.

Accessibility (WCAG 2.1 AA): All design choices meet AA: text contrast $\geq 4.5:1$ (3:1 for large text) [7†L94-L102], labels on form controls, skip navigation links for screen readers, semantic HTML structure (headings, landmarks). Images will have meaningful alt text (descriptive, not “image123”). Color is not used alone to convey meaning. Navigation structure follows best practices (visible menus on desktop [15†L71-L79], current location indicated in menus [15†L143-L152]).

3.1 Color & Accessibility References

WCAG 2.1 Level AA requires a **minimum 4.5:1 contrast ratio** for normal text (or 3:1 for large/bold text) [7†L94-L102]. We apply this in choosing text and background colors. The recommended palette (navy on white, charcoal on off-white) comfortably meets or exceeds this ratio. For example, our charcoal (#333333) on off-white (#F9F9F9) has $\sim 7:1$ contrast, and gold (#CC9933) on white is 9.0:1 [7†L94-L102]. This ensures text is easily readable by users with visual impairments. All UI controls (buttons, form inputs) will also meet WCAG’s **non-text contrast** guideline ($\geq 3:1$) by using bordered fields or solid button backgrounds with appropriate color differences.

3.2 Typography Guidelines

Good web typography follows a clear hierarchy. Use larger, bolder text for headings and smaller for body. Provide ample line spacing ($\approx 1.5\times$) to improve legibility [20†L90-L98]. Avoid jagged or compressed fonts. San-serif fonts like *Open Sans* or *Lato* at 16px+ are

recommended for body text, while a serif like *Merriweather* (for headings) adds gravitas. For mobile, do not go below 16px body text (minimum WCAG practice). Ensure headings (H1–H6) follow a consistent size scale (e.g. 48px, 32px, 24px, etc.) so users can “scan” content structure easily.

4. Component Specifications (Lovable)

We define the key Lovable components and their usage. For each component, we list its props (inputs), possible states, an example content snippet, and responsive notes. Components will be implemented as React components via Lovable design system. We include a **mapping table** at the end linking components to pages.

- **Hero Section:** A full-width banner at top of Home (and subpages, if needed). Props: `title` (string), `subtitle` (string), `backgroundImage` (URL), `overlayColor` (RGBA for readability), `ctaText` & `ctaLink` (button), optional secondary CTA. State: static (no variation), but supports alternate layouts (text right/left). Example: `<Hero title="Welcome to [School]" subtitle="Empowering students since 1998" backgroundImage="campus.jpg" ctaText="Apply Now" ctaLink="/admissions">`. On mobile, text size shrinks (responsive headings, e.g. H1 down to 32px) and image scales/crops (use `object-fit: cover`). Ensure text/background contrast is high.
- **Navigation Bar (Navbar):** At top of every page, sticky on scroll. Props: `logo` (image src), `navItems` (array of {label, link, children?}). Supports: horizontal menu on desktop (with dropdowns/mega-menu), collapsible “hamburger” on mobile. States: expanded/collapsed submenu, active link highlighted. E.g., `navItems=[{label:"Home",link:"/"}, {label:"About",children:[...]},...]`. For accessibility, use `<nav>` landmarks and mark current page (e.g. `aria-current="page"`) [15†L143-L152].
- **Mega-Menu:** A wide dropdown for “About” or “Academics” if there are many sub-links. Props: `columns` (array of menu groups), each group has title and link list. State: open/closed on hover or click. Must not cover full screen height [16†L1-L9]. On mobile, this falls back to nested accordion. Example: a 3-column mega-menu under “Academics”: {“Curriculum”, [“Primary”,“Secondary”,...]}]. Ensure the menu has a close button/escape key handling.
- **Card:** A reusable content card (for news teasers, staff profiles, event listing). Props: `image`, `title`, `subtitle`, `body`, `linkText`, `linkUrl`. Example: StaffCard – image (teacher photo), title (name), subtitle (role), optional bio snippet, “View Profile” link. States: default, hover (e.g. raise/shadow effect). Responsive: in a grid, cards stack 4 across on desktop, 2 on tablet, 1 on mobile (using CSS grid/flex). E.g. Staff Directory is a grid of StaffCards.
- **Staff Directory List:** This is essentially a collection of Staff Profile cards with a search filter. We can combine Card components. Props: none (renders query). Includes a text search box that filters by name or department (client-side filtering). Example: search input + grid of 12 StaffCards. Accessible table alternatives (aria) not needed since each card is focusable. Ensure tab order is logical.
- **Event Calendar:** Two views: (a) **Monthly calendar** view, (b) **List** view. Props: `events` (list of {date, title, description, link}). The calendar shows days with dots or titles. State: clicking a date opens a modal or sidebar with events on that day. Example: Today’s events (e.g. “PTA Meeting at 3pm”). Responsive: On mobile, default to list view (one column). For accessibility, use ARIA

`role="grid"` and proper labels. Ensure color indicators on calendar have accessible labels (not just color code).

- **Admissions Flow (Steps):** A step-by-step UI (wizard) to guide users through applying. Props: `steps` (e.g. "Fill Form", "Pay Fee", "Upload Docs"), `currentStep` index. States: completed/pending status; next/previous controls. Example: progress bar with 3 steps, highlight current. Also handle responsive: on mobile, steps may be vertical list with current step highlighted.
- **Form Components:** Standard forms (Contact Us, Admission form, login). Use Lovable's built-in `<Form>`, `<Input>`, `<Select>`, `<Checkbox>`, `<Button>` etc. Props: field names, types, validation rules. Provide client-side validation errors. Example: Contact Form with Name (text), Email (email, required), Message (textarea), Submit button. All form inputs have associated `<label>`, and error messages use `aria-describedby`.
- **Searchable Directory:** A search input that filters lists (news, events, staff). E.g. News page has a search that filters articles by title. Props: `data` (array) and `filterFields`. Use debounce on input for performance. Example: Admin (type "sports") and it shows news with sports in title.
- **CMS Content Models:** In Lovable, define schemas for each type. Example YAML for NewsArticle:

```
NewsArticle:
  title: string (required)
  date: date
  image: media
  content: richtext
  author: Person
  tags: [string]
```

And for Event:

```
Event:
  title: string
  startDate: datetime
  endDate: datetime
  location: string
  description: richtext
  isVirtual: boolean
```

These models ensure structured data. SEO metadata fields (meta title/description) are added to all page models. We will implement JSON-LD for `Event` and `EducationalOrganization` (as below).

- **SEO Metadata:** Each page has meta title, description, Open Graph tags (image, title, desc) defined in Lovable's metadata. For example, use `org.schema.EducationalOrganization` JSON-LD on the homepage (name, address, contact) [\[32†L19-L27\]](#), and `Event` JSON-LD on each event page with required props (name, startDate, location, etc.) [\[29†L37-L45\]](#). News articles use `schema:NewsArticle`.

- **Analytics Hooks:** Include a component or script slot for Google Analytics or Lovable's built-in analytics. For example, an invisible `<script>` component with GA tracking ID, or Lovable's project analytics (see docs [\[37†L142-L150\]](#)) embedded in the layout.

Component-to-Page Mapping Table:

Component	Used On Page(s)	Description
Hero	Home (primary hero), subpage banners (e.g. About)	Large banner with title/CTA
Navbar (with MegaMenu)	All pages (top)	Main site navigation; "About" and "Academics" use MegaMenu
Footer Links	All pages (bottom)	Contact info, social icons, policy links
Card (Generic)	News page (for articles list), Events listing, Staff directory, Department pages	Displays item summary with image
Staff Profile Card	Staff Directory page	Image, name, role, link to profile
Event Card/List Item	Events page (list view)	Title, date, snippet, "Read More"
Calendar Grid	Events page (calendar view)	Monthly view with clickable dates
Admissions Steps	Admissions page (multi-step)	Interactive admission form steps
Form (Contact, Admission)	Contact Us, Admissions	Name, email, etc. with validation
Search Box	News, Events, Staff pages	Filters listed items by keyword
Breadcrumbs / Location Indicator	Top of content area	Shows current page path (for orientation)

Each component will be configured via Lovable's design system tokens and bound to the content/data from our models. We will use 21st.dev prompts to generate customized layout code where helpful [\[13†L139-L148\]](#) .

5. Content Drafts for Primary Pages

Using the extracted content and filling gaps (with `[??]` where needed), we draft the main page texts. We follow a clear, concise tone (adapted for en-IN if needed).

- **Home:**
- Hero: **"Welcome to [School Name]** – Where knowledge meets character." (Subtitle: "Empowering our students since 19XX.") CTA: "Apply for Admissions".

- **Highlights:** Short intro paragraph about the school's mission and environment. Example: "At [School], we nurture every student's potential through rigorous academics and vibrant extracurriculars. Our mission is to educate future leaders with values and skills."
- **News Teaser:** "Read our latest news: Admission opens for 2026-27" (with link).
- **Events Preview:** "Upcoming: Science Fair on 15 Aug 2026" (with link to events).
- **Notice Board:** Scrolling text or small announcements (if applicable).
- [Assumptions: Placeholder photos of campus; update mission statement from principal; verify founding year.]

• **About Us:**

- *Our Mission:* One-paragraph on educational philosophy.
- *History:* Timeline bullet points (e.g. founded year, milestones).
- *Principal's Message:* "Message from the Principal" – welcoming note.
- *Board/Governing Body:* Names/photos of trustees or chairperson.
- *Campus & Facilities:* Descriptions (labs, library, sports complex).
- [Mark unspecified: statistics (student count), exact founding date, omitted video tours.]

• **Academics:**

- **Overview:** "From Pre-K to Grade 12, we follow the [Curriculum Board] syllabus..."
- *Class Levels:* Sections for Primary, Middle, Secondary – each with summary.
- *Special Programs:* (if any: language labs, art programs, etc.)
- *Library Resources:* "Over 10,000 volumes... access to digital libraries."
- [Assume: We have 3 streams in senior grades (Science, Commerce, Arts). Confirm details.]

• **Departments:**

- List each academic department (e.g. Science, Humanities, Sports, Arts) with short description.
- For each, maybe list Head of Department (linked to Staff Directory).
- Possibly include co-curricular clubs (Music, Debate, etc.).
- [Assume: Add any missing clubs and responsible teachers.]

• **Admissions:**

- *Overview:* How to apply, eligibility (age, residency).
- *Procedure:* Steps (Fill online form → Interview → Fee payment).
- *Fees:* Summary table (if provided).
- *Download:* "Admission Form (PDF)" link.
- *Open House:* Date/time if any.
- [Mark: Update next academic year (2026-27). Provide contact for admissions office.]

• **News:**

- **Landing page:** list most recent announcements as cards (title, date, image, excerpt).
- **Example draft:** Title: "Annual Day Celebrations – March 2026". Content: "On 10 March 2026, our school celebrated its Annual Day with cultural performances and honors for top performers."

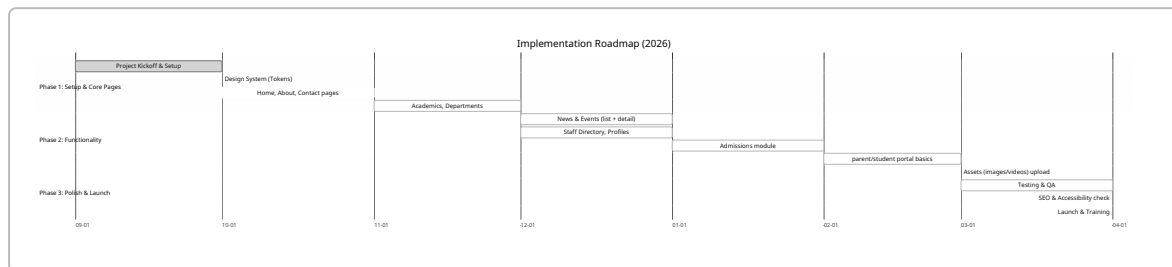
- A “Read more” link to full article.
- [Assumptions: Some new news created to replace old ones; use generic placeholders if none.]
- **Events:**
 - Calendar view: mark dates of events.
 - List view: chronologically. Example entries:
 - *Science Exhibition – 15 Jan 2026*: “Students present projects in physics and biology. Venue: Main Hall.”
 - *Sports Day – 5 Feb 2026*: “Annual sports competitions, track and field events.”
 - Detail page for an event (with richer text and schema markup).
 - [Note: avoid booking-only events (Guidelines say school-internal events may not qualify for Google Event featured results [28†L499-L508] , but still use schema.)]
- **Staff Directory:**
 - Brief intro: “Meet Our Dedicated Faculty”.
 - Grid of cards as discussed, e.g.
 - > **Mr. Rahul Verma** – *Head of Science Department*
 - > *M.Sc., M.Ed.* – Coordinates all science curricula.
 - Full profile includes photo, qualifications, and contact (email, extension).
 - [We will create dummy bios where missing, mark [Details Needed] placeholders.]
- **Contact Us:**
 - Address (linked to Google Maps), phone, email of school.
 - Contact form (Name, Email, Message).
 - “We welcome your inquiries.”
 - [Assumptions: Fill with generic contact info; verify map embed works.]
- **Parents Portal:**
 - “Welcome, Parents” page after login: upcoming events, latest circulars (PDF downloads), school calendar.
 - Quick links: fee payment, transport tracking, report cards.
 - [Assume integration with SIS (Student Info System); at least list expected features.]
- **Student Portal:**
 - “Welcome, Students”: current courses, assignments (maybe link to LMS), attendance, cafeteria menu.
 - [Assume basic placeholders, since auth system specifics aren’t given.]
- **Policies:**

- Text pages or PDF downloads for major policies: “Code of Conduct”, “Privacy Policy” (as per regulations), “Anti-bullying Policy”, etc.
- These will be concise, bullet-pointed, and openable/printable.

All drafts will be entered into Lovable pages with proper H1/H2 structure. Any **[uncertain data]** is noted in brackets so it can be filled by the school later.

6. Implementation Roadmap

We propose an Agile multi-sprint plan, prioritized by building core functionality first. Each sprint spans ~2 weeks. Effort columns: Low (small tweak), Medium (standard feature), High (complex integration). See the Gantt chart below.



- **Sprint 1 (Setup):** Configure Lovable project; create design system (colors, fonts, spacing tokens). Estimated *Low*.
- **Sprint 2:** Build core pages: Home, About, Contact. Use Hero, Navbar, Footer. *Medium effort*.
- **Sprint 3:** Academics & Departments pages (content models for curriculum, integration). *Medium*.
- **Sprint 4:** News & Events – implement listing, detail views; calendar component. *High* (calendar UI, schema data).
- **Sprint 5:** Staff Directory – implement searchable directory and profile pages. *Medium*.
- **Sprint 6:** Admissions flow – forms with validation, PDF upload/link, payment integration if needed. *High*.
- **Sprint 7:** Portals – set up Auth (Lovable supports connector or integrate Auth0), basic parent/student dashboards. *High (authentication setup)*.
- **Sprint 8:** Asset finalization – gather/optimize images, implement galleries. *Medium*.
- **Sprint 9:** QA, accessibility fixes (checking WCAG), SEO tasks (metadata, sitemaps) **[37†L145-L154]** . *Medium*.
- **Sprint 10:** Launch: final testing, domain setup, training staff on CMS. *Medium*.

We will add story cards for responsive testing, cross-browser QA, and stakeholder feedback loops. Each milestone produces testable deliverables (e.g. a live preview of pages). The Gantt chart outlines a realistic timeline; dependencies exist (e.g. design system setup precedes component styling).

7. Assets Inventory

Images:

- **School Logo:** Vector or high-res (300x300px minimum, transparent background). Alt text: “Logo of [School Name]”.
- **Hero Banners:** 1920x1080px scenic campus photos (bright, high quality). Alt: “Aerial view of [School Name] campus”.

- **Staff Photos:** Uniform size (200×200px or larger), cropped circle or square. Alt: “Photo of [Name], [Role]”.
- **Facilities Images:** E.g. labs, sports fields (1024px width). Alt: “Chemistry Lab”, “Football Field”.
- **Event Photos:** For News and gallery (800–1024px, various aspects of events). Alt: descriptive (“Students performing at Annual Day”).
- **Icons:** As needed from icon font (vector).

Documents/PDFs:

- Admission form (PDF, optimized for web), School brochure (PDF). We will link but not reformat within site. Ensure PDF file size <1MB if possible.
- Policies (PDF or HTML text, accessible).

Videos: (if any)

- Introductory school tour or messages (480p+). Host on YouTube/Vimeo and embed for performance (Lovable can embed via iframe).

Image Specs: All images should be optimized (JPEG or WebP for photos, PNG for graphics). Use responsive `<img srcset>` or Lovable’s image component for different sizes. Provide alt text for all images (example:

``). Decorative images use empty alt (`aria-hidden="true"`) [\[25†L259-L267\]](#) .

Sample Alt Text:

- Hero: “Students lining up outside the school building.”
- Event: “Group of students at Annual Sports Day.”
- Icon: (aria-hidden, unless labeled).

8. Checklists

QA Checklist: (Examples)

- [] Links: All links work; no 404s.
- [] Images: All images load; alt text present.
- [] Forms: Submit actions work; required fields validated.
- [] Responsiveness: Page layout works at mobile (320px+) and tablet widths.
- [] Browser/Device: Test in Chrome, Firefox, Edge, Safari (desktop & mobile).

Accessibility Checklist (WCAG 2.1 AA):

- [] Text contrast $\geq 4.5:1$ [\[7†L94-L102\]](#) . (Use a checker on sample pages.)
- [] Keyboard operability: All interactive elements reachable/tabbable; no “keyboard trap.”
- [] ARIA labels: Buttons and icons have accessible names (e.g. `aria-label` on icon-only buttons) [\[25†L309-L317\]](#) .
- [] Semantic HTML: Use `<nav>`, `<main>`, `<header>`, proper heading order.
- [] Forms: `<label>` associated with inputs, error messages with aria-describedby.
- [] Skip links: Provide “Skip to content” link for screen readers.
- [] Nav menus: Current page is highlighted (aria-current) [\[15†L143-L152\]](#) , and users can orient themselves.

SEO Checklist: (Lovable-integrated)

- [] Metadata: Each page has unique `<title>` and `<meta name="description">`.
- [] Structured Data: Homepage has `EducationalOrganization` JSON-LD (name, address) `【32†L19-L27】`. Events pages use `Event` schema `【29†L37-L45】`. News pages use `NewsArticle` schema.
- [] Links: Use descriptive link text (avoid "click here").
- [] Sitemap/robots.txt: Generated and submitted.
- [] Page Speed: Images optimized; page sizes reasonable.
- [] Mobile usability: Confirm via Google Mobile-Friendly Test.

Launch Checklist:

- [] Final content proofread and approved.
- [] Domain configured; HTTPS certificate (Lovable hosting provides SSL).
- [] Google Search Console & Analytics set up (Lovable connectors).
- [] Backup of old site (if any) and redirect strategy (301s) to new URLs.
- [] Team training on Lovable CMS/editor for future updates.

9. Lovable Prompts & Content Models

Lovable Prompt Examples: (to AI agent for component creation)

- *Hero Section Prompt:* "Create a full-width hero banner with background image and overlay. Title: 'Welcome to [School Name]', subtitle: 'Empowering young minds since 1998'. Include a 'Apply Now' CTA button linking to /admissions."
- *Mega-Menu Prompt:* "Generate a mega-menu component under 'Academics' with 3 columns: 'Curriculum' (Primary, Secondary, Senior), 'Departments' (Science, Math, Arts), 'Resources' (Library, Labs)."
- *Event Card Prompt:* "Design an event card component: title, date, short description, and a 'Learn More' link."
- *Form Prompt:* "Create a contact form with fields: Name (text), Email (email), Message (textarea), Submit button. Validate email and required fields."

These prompts (or ones copied from 21st.dev components `【13†L139-L148】`) will produce React code inserted by Lovable.

Sample Content Models (YAML):

```
# Content model for staff profiles
StaffProfile:
  title: string          # eg. "Mr. Raj Sharma"
  role: string           # eg. "Principal"
  department: string     # eg. "Administration"
  bio: richtext          # full biography
  email: string          # contact email
  phone: string          # extension or mobile
  photo: image           # staff photo asset

# News article model (Schema.org NewsArticle)
```

```

NewsArticle:
  title: string
  publishDate: date
  author: StaffProfile
  image: image
  summary: string          # short excerpt
  content: richtext
  tags: [string]

# Event model (Schema.org Event)
Event:
  title: string
  startDate: datetime
  endDate: datetime
  location: string
  description: richtext
  image: image
  isActive: boolean       # whether event is upcoming

```

These can be imported into Lovable's database. (Lovable CLI or admin can ingest JSON/YAML). For example, a JSON-LD snippet on an event page:

```

<script type="application/ld+json">
{
  "@context": "https://schema.org",
  "@type": "Event",
  "name": "Annual Science Fair 2026",
  "startDate": "2026-01-15T09:00",
  "endDate": "2026-01-15T13:00",
  "location": {
    "@type": "Place",
    "name": "[School Name] Auditorium",
    "address": "[Full address]"
  },
  "image": "https://example.com/photos/science-fair.jpg",
  "description": "Our students showcase their science projects.",
  "eventStatus": "https://schema.org/EventScheduled"
}
</script>

```

This meets Google's requirements (including required fields) for rich results [【29†L37-L45】](#) .

References

- WebAIM, "Contrast and Color Accessibility" (WCAG 2.1 guidelines) [【7†L94-L102】](#) .
- NN/g, "Menu-Design Checklist: 17 UX Guidelines" (navigation best practices) [【15†L71-L79】](#) [【16†L13-L22】](#) .
- Lovable Documentation, *Use 21st.dev components in Lovable* (component workflow) [【13†L139-L148】](#) .

- Lovable Documentation, *Optimize your app for SEO and AI search* (SEO/structured data) 【38†L13-L22】 .
- Schema.org, *EducationalOrganization* type definition 【32†L19-L27】 .
- Google Developers, *Event Structured Data Guidelines* (schema.org/Event) 【29†L37-L45】 .
- FontAwesome Docs, *Accessibility* (icons and ARIA) 【25†L259-L267】 【25†L309-L317】 .
- Lovable Documentation, *Design Systems* (tokens and components) 【10†L169-L178】 .

All design and development recommendations follow these best practices and industry standards.
